



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

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Describe the problem statement

When a hospital emergency department is overcrowded, deciding which patient should be seen first depends on a nurse's judgment. A mistake or delay can have serious consequences. A nurse may manage tasks at once, such as triaging patients, arranging transport, and watching the waiting room for signs of deterioration. This may happen within the first two minutes of arrival, even when information is missing, such as vital signs or EHR access, or when a language barrier makes communication difficult.

Research suggests that nearly one in three emergency department (ED) visits may be assigned the wrong level of urgency, while only about two out of three critically ill patients are correctly identified when they arrive. The situation can be more challenging in many Indian public hospitals, where formal triage systems may be limited during busy periods.

This is not necessarily a problem with nurses or their decision-making. The bigger issue is how the current system works. Most triage methods assess patients at arrival and then move on. They often do not continuously check patients while they wait. As a result, a patient whose condition worsens may not be noticed until it becomes serious.

What is needed: An AI-powered decision-support assistant that supports staff from the moment a patient arrives. It can help prioritize and direct patients while monitoring those waiting for signs of worsening. The goal is to reduce waiting times and under-triage without replacing clinical judgment. It should work reliably in low-resource, high-surge, multilingual emergency departments, even without EHR access.

Proposed solution

Acuity is proposed as an AI-powered decision support system designed for the ED entrance and waiting room. It is not a diagnostic chatbot and does not replace the triage nurse. Instead, it supports nurses by helping them make faster and more consistent decisions. It works through three main layers:

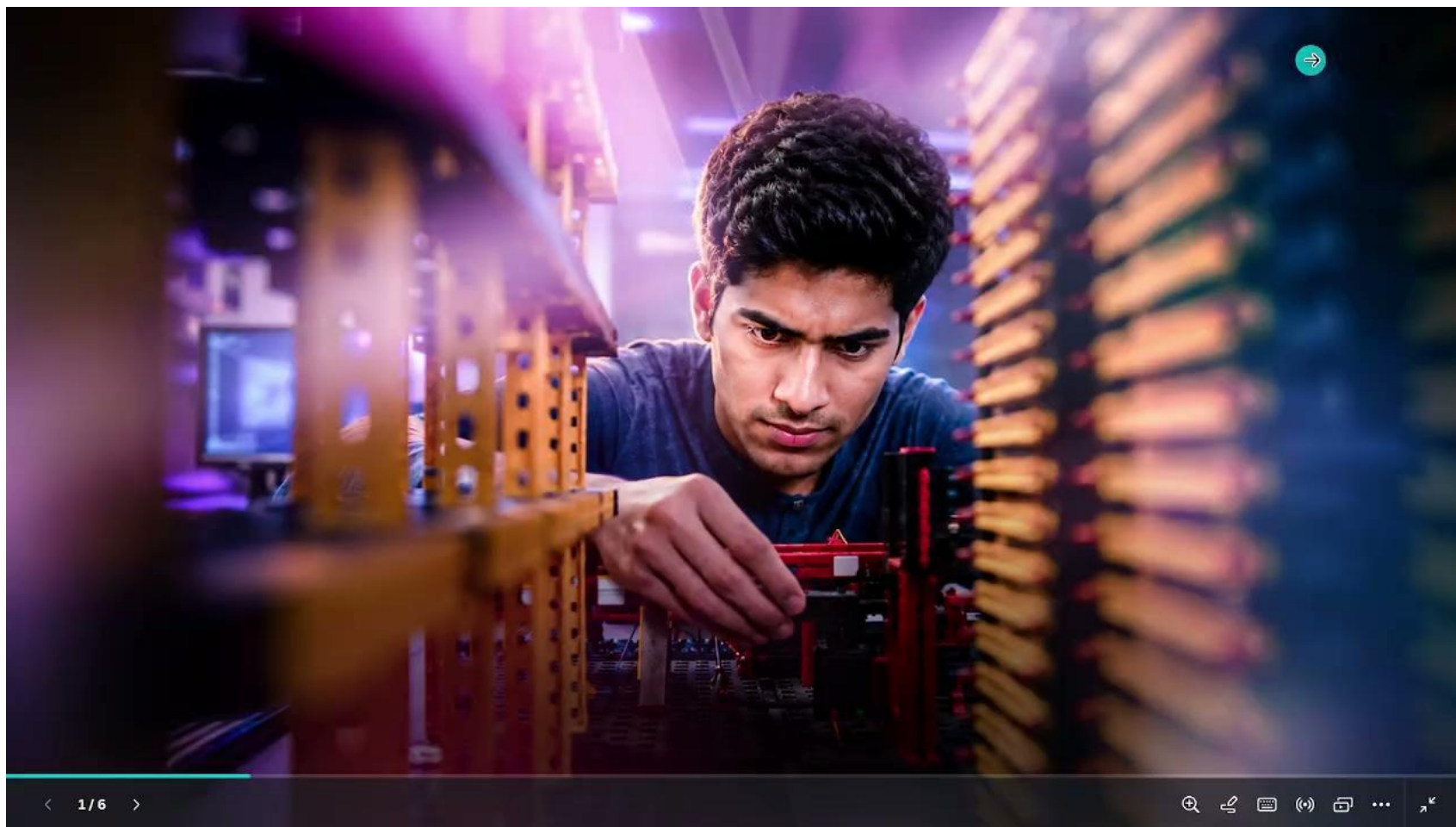
DOOR (0–90 sec): Acuity will use whatever information is available, such as visible signs, the patient's complaint, and any recorded vital signs, to recommend an acuity level (Red/Yellow/Green, mapped to ESI). If any information is missing, the system will never assume that everything is normal. Instead, missing data will increase uncertainty and may lead to a higher recommended acuity.

FLOW (live): The system will help direct patients to the appropriate area, such as emergency care or fast-track, based on the available bed capacity. Within each priority level, it will maintain a fair first-in-first-out queue, helping reduce bias.

WATCH (until seen): Acuity will continue to monitor patients while they wait. If someone's condition appears to worsen, the system will automatically raise an alert for reassessment. This will make triage an ongoing process rather than a one-time decision when the patient arrives.

Nurse remains in control: The nurse will always have the final say on acuity, diagnosis, and any downgrade. Acuity will mainly handle operational decisions such as routing, queue order, and when a patient should be reassessed. In a genuine emergency, it can automatically raise an alert, but the nurse can override it. It will never automatically downgrade a patient and will be designed to fail safely, even during surges, with incomplete information or without EHR access. EHR access.

Video



Thank you

